Marketing Campaign Analysis

Challenge-Based Trainer Guide | Python Data Analysis | 2 Hours

Dataset note: synthetic campaign records for classroom use. Validate findings against ad-platform exports before making real budget decisions.

1. Business scenario

A marketing team manages campaigns across Facebook, Google, Instagram, TikTok, and Email. Management wants to understand which channels generate efficient traffic and conversions, which audiences have lower acquisition costs, and where campaign spend may be wasted. Students work in teams and receive hints progressively rather than a full solution at the start.

2. Challenge outputs

• A data-quality report covering missing values, duplicates, invalid negatives, and extreme values.

• A channel-level KPI table: spend, impressions, clicks, conversions, CTR, CVR, CPC, CPA, revenue, and ROAS.

• One Matplotlib chart, one Seaborn chart, and one interactive Plotly chart.

• A Streamlit dashboard with filters and KPI cards.

• Three evidence-backed findings and two marketing recommendations.

3. Two-hour challenge schedule

Time	Challenge	Hint / checkpoint
0–10 min	Understand the business questions and campaign objectives.	Awareness, traffic, leads, and conversion campaigns need different success measures.
10–25 min	Import and profile the dataset.	Try shape, head, info, isna().sum(), duplicated().sum().
25–45 min	Clean and validate the data.	Negative spend/clicks/conversions are invalid; investigate extreme positive values.
45–70 min	Engineer KPIs and aggregate by channel/audience.	Use ratios of totals for CTR, CVR, CPA, and ROAS.
70–90 min	Build charts to answer the questions.	Bar chart for ROAS; boxplot for spend spread; scatter for CTR vs CVR.
90–115 min	Run the Streamlit dashboard.	Compare channel, campaign type, region, and audience with filters.
115–120 min	Present one finding and one action.	Mention metric, sample size/volume, limitation, and next test.

4. KPI definitions

KPI	Formula	Interpretation
CTR	Total clicks ÷ total impressions	Traffic engagement; use aggregate totals.
Conversion rate	Total conversions ÷ total clicks	Share of clicks that convert; define conversion consistently.
CPC	Total ad spend ÷ total clicks	Average cost per click.
CPA	Total ad spend ÷ total conversions	Average cost per acquisition; undefined if conversions = 0.
ROAS	Attributed revenue ÷ ad spend	Revenue return, not profit; ignores product margin unless modeled.

5. Guided code checkpoints

Import and inspect

```
raw = pd.read_csv("marketing_campaign_raw.csv")
print(raw.shape)
print(raw.head())
print(raw.info())
print(raw.isna().sum())
print("Duplicates:", raw.duplicated().sum())
```

NumPy: use median and percentiles to understand skew and unusual spend.

```
spend = pd.to_numeric(raw["Spend_EGP"], errors="coerce").dropna().to_numpy()
print(np.median(spend), np.percentile(spend, 95))
```

Feature engineering hint

```
df["CTR"] = np.where(df["Impressions"] > 0,
                     df["Clicks"] / df["Impressions"], np.nan)
df["Conversion_Rate"] = np.where(df["Clicks"] > 0,
```

```
                    df["Conversions"] / df["Clicks"], np.nan)
df["CPC_EGP"] = np.where(df["Clicks"] > 0,
                    df["Spend_EGP"] / df["Clicks"], np.nan)
df["CPA_EGP"] = np.where(df["Conversions"] > 0,
                    df["Spend_EGP"] / df["Conversions"], np.nan)
df["ROAS"] = np.where(df["Spend_EGP"] > 0,
                    df["Revenue_EGP"] / df["Spend_EGP"], np.nan)
```

6. Visualization and library mapping

• NumPy: median and percentiles for spend; safe numerical calculations.

• Pandas: data cleaning, feature engineering, groupby/agg, monthly trends.

• Matplotlib: static ROAS-by-channel bar chart.

• Seaborn: spend distribution by campaign type using a boxplot.

• Plotly: interactive CTR-versus-conversion-rate scatter plot.

• Streamlit: filters, KPI cards, monthly trends, channel and audience comparisons.

7. Discussion prompts and answer cues

Which channel is best? No single answer: compare ROAS, CPA, conversion volume, and campaign objective. Email may have low spend and a small volume.

Should we pause high-CPA campaigns? Check attribution, conversion quality, spend level, and objective first; consider a controlled test.

Why use ratios of totals? Averages of campaign-level ratios can give tiny campaigns the same weight as large campaigns.

Does high CTR mean campaign success? Not necessarily. High CTR with low conversion may indicate audience, offer, or landing-page mismatch.

What does ROAS mean? Revenue divided by ad spend; it is not profit unless margins and other costs are considered.

8. Dashboard run instructions

• Install: pip install -r requirements.txt

• Run: streamlit run app.py

• Keep marketing_campaign_raw.csv beside app.py.

• Demonstrate channel, campaign type, region, and audience filters.

Instructor note: provide the raw CSV and challenge questions first. Reveal hints progressively. Use the starter script and dashboard as a reference/checkpoint after students attempt the work.